



SECURITY ESSENTIALS

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ATTACKERS



Attackers are creative and will attempt many different techniques to trick users

WHAT IS “ATTACK “

- Any attempt to :
 - destroy, expose, alter, disable, steal or breaking into the information
 - breaking the systems.
 - gain unauthorized access to or make unauthorized use of an asset
- Could be intentional or unintentional



CATEGORIES OF ATTACKS

- **Passive Attack**

- Difficult to detect, because the attacker **isn't actively** sending traffic (malicious or otherwise)
- Example: An attacker capturing packets from the network and attempting to decrypt them



- **Active Attack**

- Easier to detect, because the attacker **is actively** sending traffic that can be detected.
- An attacker might launch an active attack in an attempt to access information or to modify data on a system.



ATTACKERS TYPES

- **Script kiddy**

- A script kiddie, or “skiddie,” is someone who lacks programming knowledge and uses existing software(downloads hacking utilities) to launch an attack. Often a script kiddie will use these programs without even knowing how they work or what they do.



ATTACKERS TYPES

- **Hacker**

- A person who enjoys learning the details of computer systems and stretch their capabilities.



ATTACKERS TYPES

- **Cracker (Black Hats)**

- A person who uses his hacking skills for **Offensive purposes**. (destructive activities)

- **Ethical hacker (Security Analysts, White Hats)**

- Security professionals who apply their hacking skills for defensive purposes.

- **Gray Hats**

- Individuals who work **both Offensively and Defensively** at various times.



ATTACKERS TYPES

- **Hacktivist**

- A hacker with **political motivations**, such as someone who defaces the website of a political candidate
- intend to hack **government websites**.
- They pose themselves as activists,
- Hacktivist can be an individual or a bunch of nameless
- The **data** gained from government files accessed **are used for personal political or social gain**.



ATTACKERS TYPES

- **Phreaker**
 - A phreaker is a hacker of a telecommunications system.



ATTACKERS TYPES

- **State/Nation Sponsored Hackers**
 - Government appoints hackers to use their knowledge to **gain information about / from other countries.**
 - Sensitive information aids to be on **top of every situation and to avoid upcoming danger to their country**
 - They report only to their governments.



REMEMBER

Script kiddie



An unskilled hacker who compromises system by running scripts, tools and software developed by real hackers

Cyber Terrorist



Individuals with wide range of skills motivated by religious or political beliefs to create fear by large-scale disruption of computer networks

State sponsored Hacker



Individuals employed by the government to penetrate and gain top-secret information and to damage information systems of other governments

Hacktivist



Individuals who promote a political agenda by hacking, especially defacing or disabling websites

WHAT DOES A MALICIOUS HACKER DO?



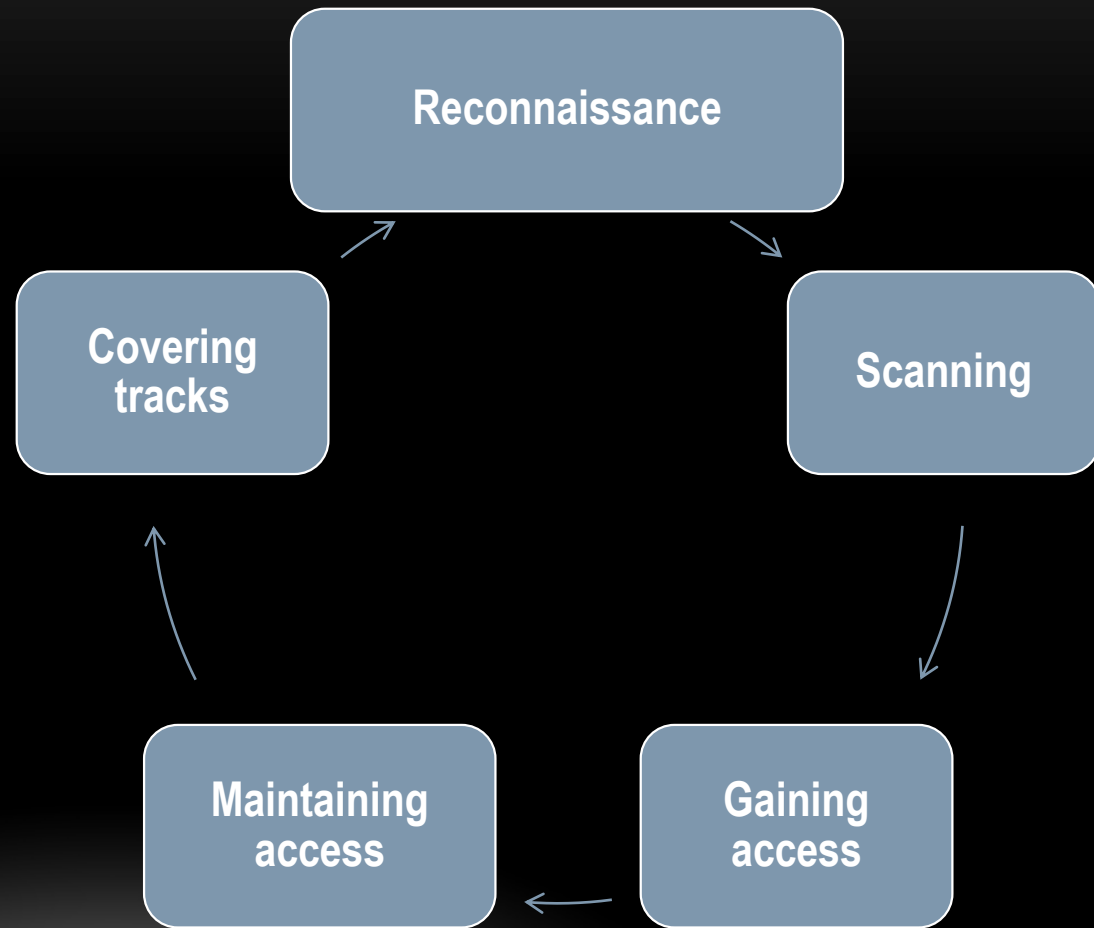
TRY TO THINK AS A HACKER

- You are A hacker
and your task to thinking
of how you can hacking
the web server of
zeko company
their website is
www.zeko.com
- And they have an
administrator
the webserver is hosted
at their company



5 STEPS

- Reconnaissance
- Scanning
- Gaining access
- Maintaining access
- Covering tracks



PHASE 1 - RECONNAISSANCE

- Reconnaissance

- refers to the preparatory where an attacker seeks to **gather as much information as possible about a target** to launching an attack.
- Gathering info about internal structure of organization, by browsing and search the internet



RECONNAISSANCE (COLLECTING INFORMATION)

- **Resources**

- Search engines

- Ask Google ?



- Social media

- Facebook, twitter, Instagram, etc.



- Forum

- Ex. Microsoft ----> <https://answers.microsoft.com/>

- History

- <http://archive.org/>

- Domain name, mail messages, news groups posts and web servers pages

- Nslookup command

- Live camera

<http://www.insecam.org/en/bycountry/EG/>

LAB

<https://answers.microsoft.com/>

← → ↺ 🏠

answers.microsoft.com/ar-sa/windows/forum?sort=LastReplyDate&dir=Desc&tab=All&status=all&mod=&modAge=&advFil=&postedAfter=&postedB...

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إظهار الكل

إظهار الكل

Windows

↓ إظهار عوامل التصفية فرز حسب الأحدث

الكل

9

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9

مرات العرض

مفيد

الردود

[Smug Puppy](#) بواسطة [mariamelia1](#) | تم إنشاء ذو القعدة 1442, 5 بواسطة

مساء الخير، انا تواجه مشكلة ان سطوع شاشة اللابتوب عندما لا يكون على الشاحن يتغير سطوع الشاشة بطريقة تلقائية و تعتم الصفحة عند الدخول لصفحة ذات خلفية داكنة لقد قمت بايقاف عرض تكنولوجيا توفير الطاقة من اعدادات تحكم رسومات Intel

1 بحاجة إلى إجابة

3

0

0

مرات العرض

مفيد

الردود

[mahamed khalloug](#) بواسطة [mahamed khalloug](#) | تم إنشاء منذ 6 ساعة 7 دقيقة بواسطة

السلام عليكم،ارجوا المساعدة في حل مشكلة ايقاف تحديثات ويندوز 7 وشكرا.محمد خلوق.

2 بحاجة إلى إجابة

70

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3

مرات العرض

مفيد

الردود

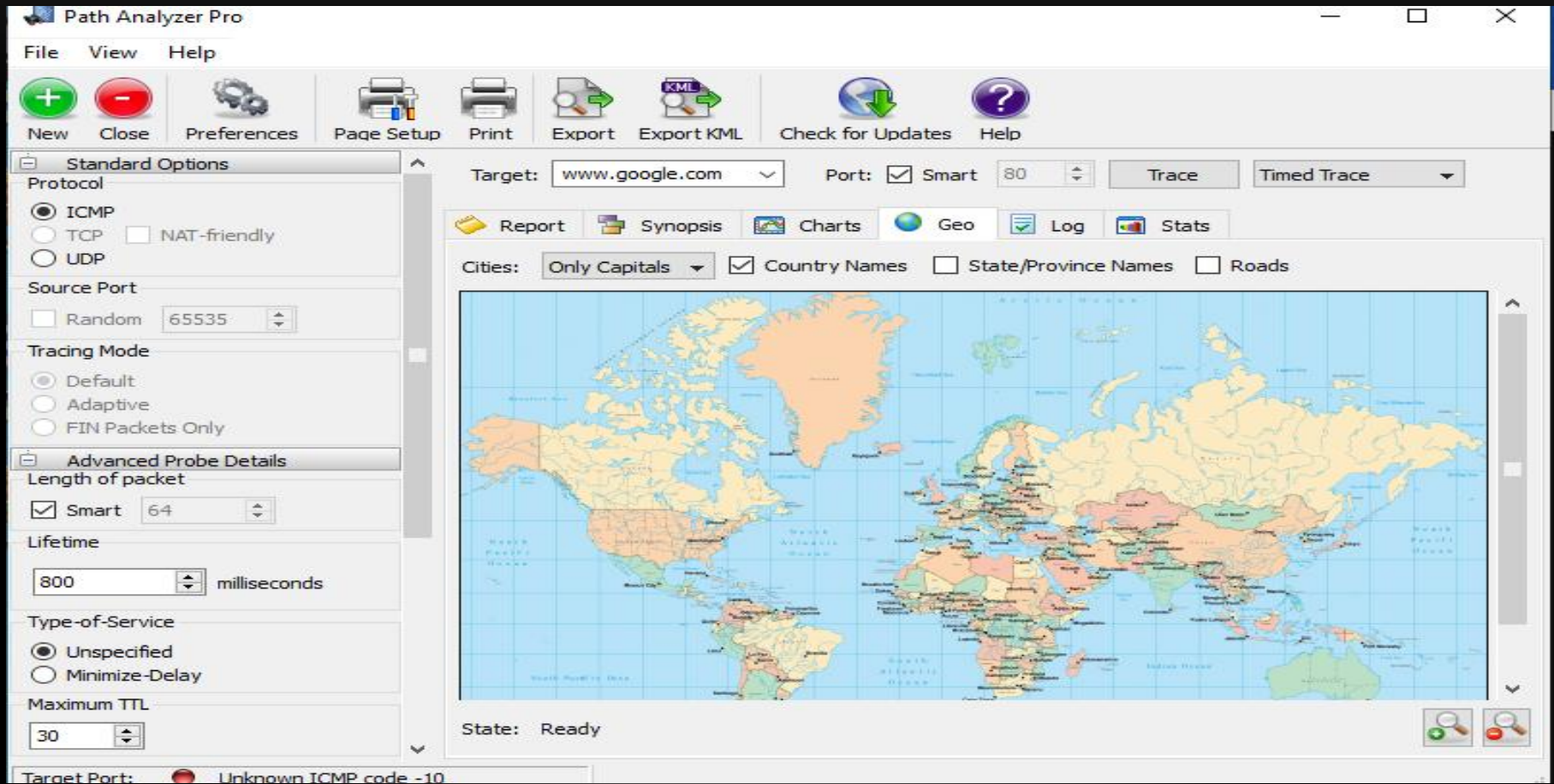
[MahmoudElmondj](#) بواسطة [trade111r](#) | تم إنشاء جمادى الثانية 1442, 25 بواسطة

لا استطيع تثبيت ويندوز 7 على جهاز لينوفو 520 حيث عند بدأ التنصيب تتوقف التاتش باد والكيبورد سواء كان التنصيب من الفلاشة أو الاسطوانة، اما عند تثبيت ويندوز 10 لا توجد مشكلة، لكنى اريد ويندوز 7

1 بحاجة إلى إجابة

- Trace route command
 - Trace the network path from one host to another
 - Tracert command
 - [Tracert zeko.com](#)

PATH ANALYZER PRO (TRACERT)



PHASE 2 - SCANNING

- Refers to pre-attack phase
- hacker scans the network with specific information gathered during reconnaissance.
- Scanning for open ports, operating systems, applications, open shares,

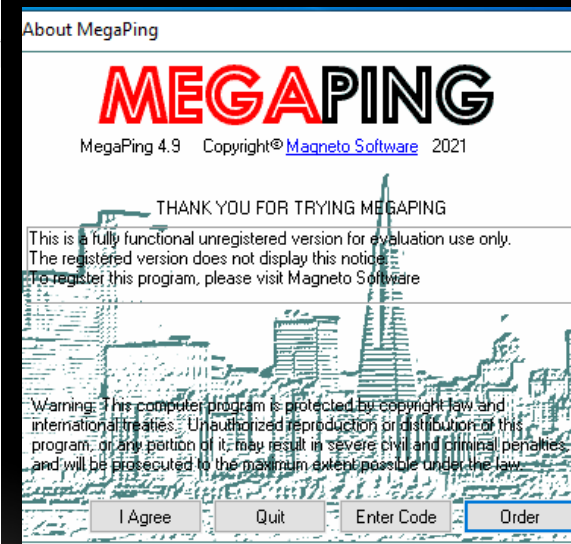
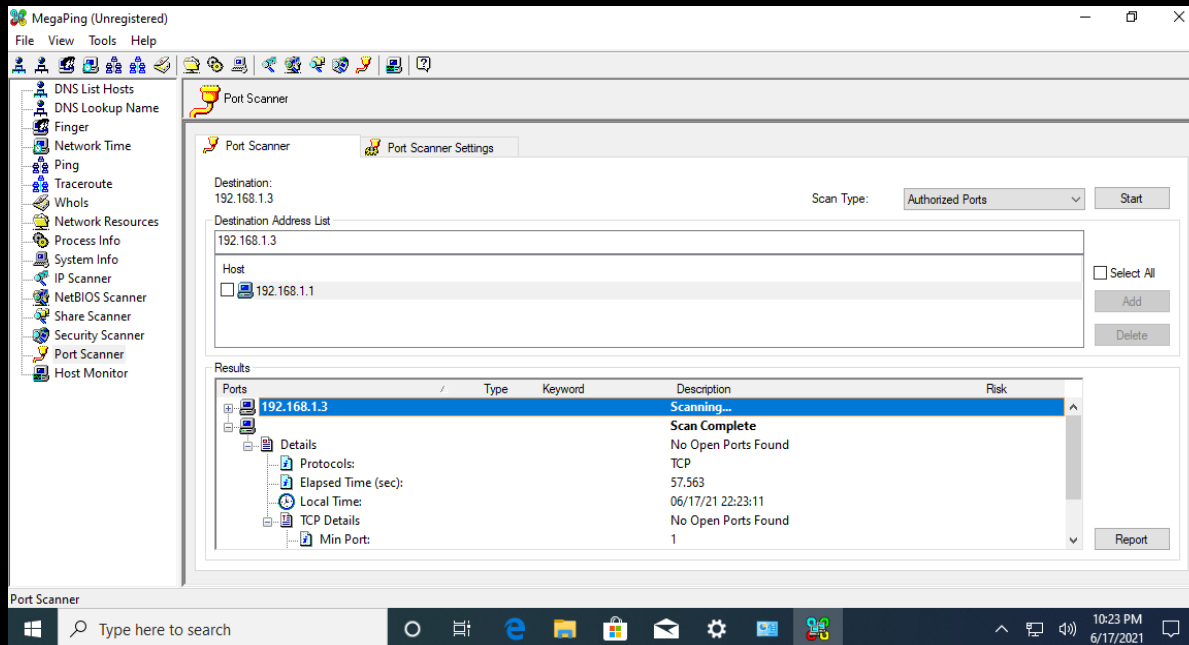
PROBING THE NETWORK

- Host and service scanning
 - Find every system on the network
 - Find every service running on each system
 - Find each exploits each service is vulnerable to
 - Ex:<https://www.acunetix.com>



LAB 4

- Port scanner
- Mega Ping
- NetScan Tools Pro Demo



PHASE 3 - GAINING ACCESS

- Refers to the true attack phase.
- The **hacker exploits the system**.
- The exploit can occur over a LAN, locally, Internet.
- Examples include buffer overflows, denial of service, session hijacking etc.

LAB 5 : LAUNCHING THE ATTACK

- Using your Hacking tools or commands on Vulnerabilities
- Tools used depend on your attack type will discussed later in Common Attacks



PHASE 4 - MAINTAINING ACCESS

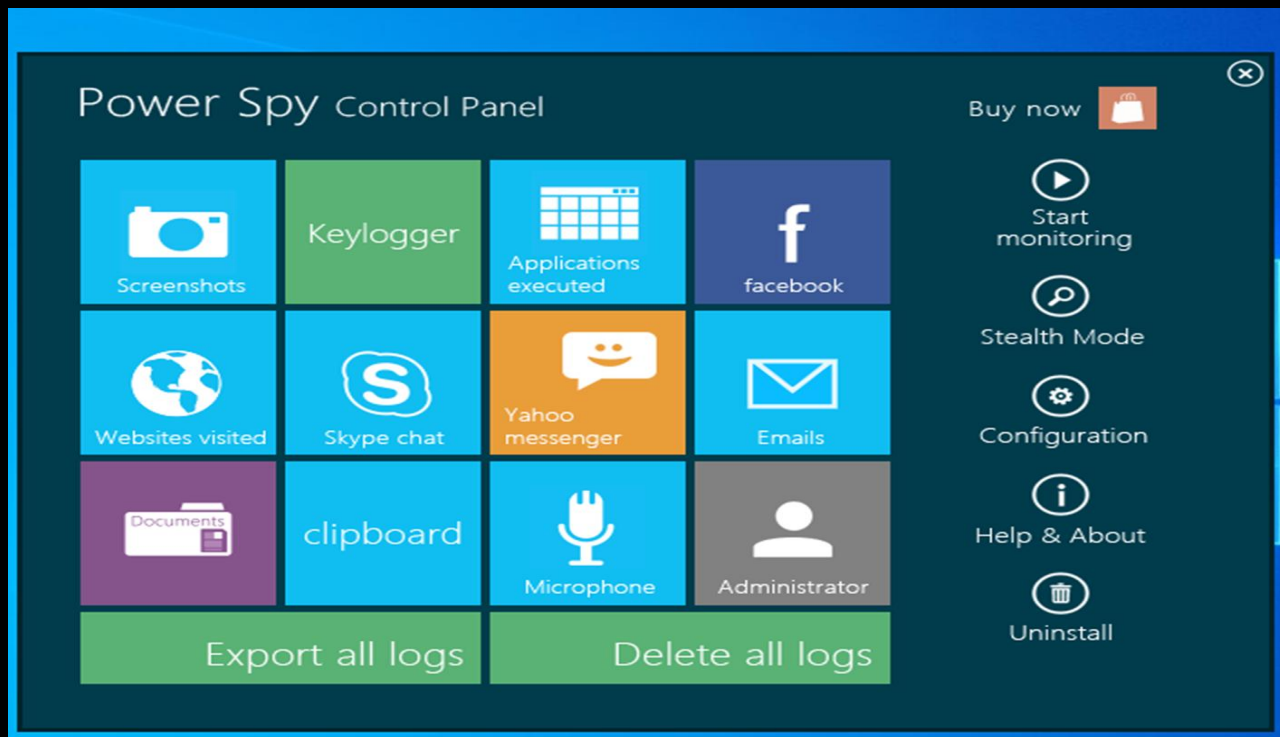
- Maintaining Access refers to the phase when the hacker tries to retain his 'ownership' of the system.
- Sometimes, hackers harden the system from other hackers as well (to own the system).
- Ex : Close the open ports

PHASE 5 - COVERING TRACKS

- Covering Tracks refers to the activities undertaken by the hacker to extend his misuse of the system without being detected.
 - Reasons include need for continued use of resources, removing evidence of hacking, avoiding legal action etc.
 - Hackers can remain undetected for long periods.
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CLEAR THE EVIDENCE

- Using your Skills to destroy evidence
- Make sure of forensics and reverse engineering



WAIT FOR THE POLICE 😊



STORY & LESSONS LEARNED





Thank You!